Faculté de Médecine

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Institut de Santé Globale

Thèse préparée sous la direction du **Professeur Olivia Keiser**

Evaluating the HIV Care and Prevention Cascade for Key Populations in Nigeria, and Building Data Science Capacity for Global Health

Évaluation de la cascade de soins et de prévention du VIH chez les populations clés au Nigéria, et renforcement des capacités en science des données pour la santé globale

Thèse présentée à la Faculté de Médecine de l'Université de Genève pour obtenir le grade de Docteur en Sciences Biomédicales, Mention Santé Globale

par

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List of Publications

Published articles

1. **Nwosu KD**, Kalaiwo A, Ngambi WF, Estill J, Omo-Emmanuel UK, Emmanuel G, Keiser O. Factors associated with viral load testing and viral suppression among HIV-positive female sex workers in Nigeria. *PLoS ONE* 2024; 19(5): e0304487. doi: 10.1371/journal.pone.0304487

2. **Nwosu KD**, Kalaiwo A, Ng'ambi W, Oyelaran O, Umoh P, Estill J, Keiser O. Oral PrEP continuation rates among key populations in Nigeria: a retrospective cohort study of a large-scale HIV prevention programme. *BMJ Global Health* 2025; 10(12): e018739. doi: 10.1136/bmjgh-2024-018739

Manuscript in preparation

3. **Nwosu KD** (co-authors to be finalized). R Basics and Beyond: Outcomes of a Cohort-Based Online Programming Course for Health Professionals. *[Target journal: Cogent Education]*.

Open-access textbooks

4. **Nwosu KD**, Vaz J, McKinley A, Valle Campos A, Merzouki A, Nguimbis B, Hsin B, Valera CB, Camara D, Araujo E, Mukonda E, Wafeu G, El Badisy I, Bensouda Korachi I, Agasa L, Nguemo L, Vancauwenberghe L, Franza M, Shrestha M, Keiser O, Rodriguez Velasquez S, Botero Mesa S. *Introduction to Data Analysis with R*. The GRAPH Courses, 2025. Available at: https://datawithr.com

5. **Nwosu KD**, McKinley A, Valle Campos A, Merzouki A, Nguimbis B, Hsin B, Valera CB, Camara D, Araujo E, Mukonda E, Wafeu G, El Badisy I, Bensouda Korachi I, Vaz J, Agasa L, Nguemo L, Vancauwenberghe L, Franza M, Shrestha M, Keiser O, Rodriguez Velasquez S, Botero Mesa S. *Introduction to Data Science with Python*. The GRAPH Courses, 2025. Available at: https://pythondatabook.com

Abstract

Background

Sub-Saharan Africa continues to bear the greatest share of the global HIV burden, and Nigeria, the most populous country on the continent, has one of the largest national epidemics in the world. Within this epidemic, key populations, including female sex workers (FSWs), men who have sex with men (MSM), people who inject drugs (PWID), and transgender individuals, face a disproportionate burden of HIV infection. Improving outcomes for these populations requires understanding where the HIV care and prevention cascades are succeeding and where gaps remain. Large-scale programmes serving key populations increasingly collect the routine data needed to answer these questions, but much of this data remains underutilized because health systems in the most affected countries often lack sufficient analytical capacity. Strengthening the HIV response therefore depends not only on generating programmatic data, but on building the local analytical capacity needed to turn that data into evidence.

Objectives

The work in this thesis pursues two complementary aims: generating programmatic evidence on the HIV cascade for key populations, and contributing to the data science capacity needed to produce such evidence. Papers 1 and 2 analyse routine programmatic data from KP-CARE 1, a large community-based HIV programme for key populations in Nigeria, examining factors associated with (a) viral load testing and suppression among HIV-positive female sex workers, and (b) oral pre-exposure prophylaxis (PrEP) continuation among HIV-negative key populations. Such analyses require data science skills that remain scarce in some of the countries most affected by HIV. Paper 3 therefore evaluates the flagship course of The GRAPH Courses, an online training platform developed to strengthen these skills among health professionals in low- and middle-income countries.

Methods

Papers 1 and 2 analysed programmatic data from Nigeria's KP-CARE 1 (Key Population Community HIV Services Action and Response) programme using mixed-effects logistic regression with random intercepts for service delivery facilities. Paper 3 analysed administrative enrollment rosters and post-course survey responses from nine cohorts of the

GRAPH Courses' R programming bootcamp, using logistic regression for quantitative outcomes and thematic coding for open-ended survey responses.

Results

The HIV programme analyses showed that among 34,976 FSWs initiating antiretroviral therapy between 2016 and 2022, 97.1% received at least one viral load test and 94.5% achieved viral suppression, with the likelihood of both outcomes improving over time. Advanced clinical stage at baseline was associated with lower odds of suppression, underscoring the importance of early diagnosis. Among 43,788 key population clients initiating oral PrEP between 2020 and 2023, the 6-month continuation rate was only 11.5%, revealing a critical gap on the prevention side of the cascade. Transgender individuals had the lowest continuation rate (3.5%), while female sex workers had the highest (13.7%).

The R course evaluation covered nine cohorts of 458 enrolled health professionals, of whom 55.9% completed the course. Among post-course survey respondents, 96% gave a recommendation score of 9 or 10 out of 10; pre-recorded video lessons and live workshop facilitator interaction were the highest-rated course components. Time constraints (34%) and competing work priorities (30%) were the most frequently cited barriers.

Conclusions

The Nigerian KP-CARE 1 programme achieved high levels of viral load testing and suppression among female sex workers, supporting the value of community-based ART delivery, but PrEP continuation was low, indicating a need for interventions that better support sustained use. The course evaluation suggests that cohort-based online training is a feasible approach to strengthening the data science capacity required to generate and use such evidence in global health programmes.

Résumé en français

Contexte

L'Afrique subsaharienne continue de supporter la plus grande part du fardeau mondial du VIH, et le Nigéria, pays le plus peuplé du continent, connaît l'une des plus grandes épidémies nationales au monde. Dans cette épidémie, les populations clés, notamment les travailleuses du sexe, les hommes ayant des rapports sexuels avec des hommes, les personnes qui s'injectent des drogues et les personnes transgenres, sont touchées de manière disproportionnée par l'infection à VIH. Améliorer les résultats pour ces populations nécessite de comprendre où les cascades de soins et de prévention du VIH fonctionnent bien et où des lacunes persistent. Les programmes à grande échelle destinés aux populations clés collectent de plus en plus les données de routine nécessaires pour répondre à ces questions, mais ces données restent souvent sous-utilisées parce que les systèmes de santé des pays les plus touchés manquent fréquemment de capacités analytiques suffisantes. Le renforcement de la riposte au VIH dépend donc non seulement de la production de données programmatiques, mais aussi du développement des capacités analytiques locales nécessaires pour transformer ces données en preuves.

Objectifs

Cette thèse poursuit deux objectifs complémentaires : produire des preuves programmatiques sur la cascade du VIH pour les populations clés et contribuer au renforcement des capacités en science des données nécessaires à la production de telles preuves. Les articles 1 et 2 analysent des données programmatiques de routine issues de KP-CARE 1, un vaste programme communautaire de lutte contre le VIH destiné aux populations clés au Nigéria, afin d'étudier les facteurs associés (a) au dépistage de la charge virale et à la suppression virale chez les travailleuses du sexe vivant avec le VIH, et (b) à la poursuite de la prophylaxie pré-exposition (PrEP) orale chez les populations clés séronégatives. De telles analyses requièrent des compétences en science des données qui restent rares dans certains des pays les plus touchés par le VIH. L'article 3 évalue donc le cours phare de The GRAPH Courses, une plateforme de formation en ligne développée pour renforcer ces compétences chez les professionnels de la santé dans les pays à revenu faible et intermédiaire.

Méthodes

Les articles 1 et 2 ont analysé des données programmatiques du programme nigérian KP-CARE 1 (Key Population Community HIV Services Action and Response) au moyen de régressions logistiques à effets mixtes, avec des intercepts aléatoires pour les structures de prestation de services. L'article 3 a analysé les listes administratives d'inscription et les réponses aux enquêtes post-formation de neuf cohortes du bootcamp de programmation R de The GRAPH Courses, en utilisant la régression logistique pour les résultats quantitatifs et un codage thématique pour les réponses ouvertes.

Résultats

Les analyses du programme VIH ont montré que, parmi 34 976 travailleuses du sexe ayant commencé un traitement antirétroviral entre 2016 et 2022, 97,1 % avaient reçu au moins un test de charge virale et 94,5 % avaient atteint la suppression virale, avec une probabilité croissante pour ces deux résultats au fil du temps. Un stade clinique avancé au début du suivi était associé à une probabilité plus faible de suppression virale, soulignant l'importance d'un diagnostic précoce. Parmi 43 788 clients des populations clés ayant commencé une PrEP orale entre 2020 et 2023, le taux de poursuite à 6 mois n'était que de 11,5 %, révélant une lacune importante du côté prévention de la cascade. Les personnes transgenres présentaient le taux de poursuite le plus faible (3,5 %), tandis que les travailleuses du sexe présentaient le taux le plus élevé (13,7 %).

L'évaluation du cours R portait sur neuf cohortes totalisant 458 professionnels de la santé inscrits, dont 55,9 % ont terminé le cours. Parmi les répondants à l'enquête post-formation, 96 % ont attribué une note de recommandation de 9 ou 10 sur 10 ; les leçons vidéo préenregistrées et l'interaction avec les animateurs lors des ateliers en direct étaient les composantes les mieux notées. Les contraintes de temps (34 %) et les priorités professionnelles concurrentes (30 %) étaient les obstacles les plus fréquemment cités.

Conclusions

Le programme nigérian KP-CARE 1 a atteint des niveaux élevés de dépistage de la charge virale et de suppression virale chez les travailleuses du sexe, ce qui soutient la valeur de la dispensation communautaire du traitement antirétroviral. Cependant, la poursuite de la PrEP était faible, indiquant la nécessité d'interventions qui soutiennent mieux son utilisation

durable. L’évaluation du cours suggère que la formation en ligne par cohortes est une approche faisable pour renforcer les capacités en science des données nécessaires à la production et à l’utilisation de ces preuves en santé globale.

Outline

This thesis is organised into the following sections:

- **General Introduction** — background on the global and Nigerian HIV epidemic, the role of key populations, the HIV care and prevention cascades, and the rationale for building data science capacity in low- and middle-income countries. The introduction concludes with the specific objectives that motivated the three studies presented in this thesis.

- **Methodological Contributions** — a description of the candidate's specific role in each of the three studies, including study design, data analysis, manuscript preparation, and (for Paper 3) curriculum and platform development.

- **Papers** — the three studies that constitute the core of this thesis, presented in full:

    - *Paper 1*: Factors associated with viral load testing and viral suppression among HIV-positive female sex workers in Nigeria (*PLoS ONE*, 2024).
    - *Paper 2*: Oral PrEP continuation rates among key populations in Nigeria (*BMJ Global Health*, 2025).
    - *Paper 3*: R Basics and Beyond: Outcomes of a Cohort-Based Online Programming Course for Health Professionals (manuscript in preparation).

- **Conclusions and Perspectives** — synthesis of the main findings across the three studies, their implications for HIV programming for key populations and for data science capacity building, and directions for future work.

- **Bibliography** — references cited throughout the thesis, in Vancouver format.

General Introduction

I. HIV PREVENTION AND CARE AMONG KEY POPULATIONS

The Global Burden of HIV and its Impact on Sub-Saharan Africa

Over four decades since its initial discovery, the human immunodeficiency virus (HIV) remains one of the most pressing global health challenges. In 2024, an estimated 1.3 million people newly acquired HIV, and 630,000 died from conditions related to acquired immunodeficiency syndrome (AIDS) (1). Although annual new infections are now down by over 60% since their peak in 1996 (1), HIV infection is chronic and lifelong, meaning that every person who acquires the virus and survives adds to the prevalent pool over time. As such, the global number of people living with HIV has continued to rise, from an estimated 26.6 million in 2000 to 40.8 million in 2024 (1). Overall, HIV/AIDS remains the fourth-largest infectious disease contributor to global disability-adjusted life year (DALY) losses, after lower respiratory infections, diarrhoeal diseases, and malaria (2,3).

Sub-Saharan Africa (SSA) bears a disproportionate share of this burden. Of the 40.8 million people worldwide living with HIV in 2024, 65% (26.3 million) were in sub-Saharan Africa, although the region contains only about 15% of the world's population (4). At its peak in the late 1990s and early 2000s, AIDS became the leading cause of adult mortality in the region; in the worst-affected countries, adult prevalence exceeded 25% and life expectancy fell by decades (4–6).

Substantial progress has been made against the epidemic since then, largely driven by the scale-up of HIV antiretroviral therapy (ART) across the region. Modern ART regimens, typically taken as daily pill formulations of multiple drugs, can suppress viral replication to undetectable levels, preventing progression to AIDS and greatly reducing the risk of HIV transmission to others (7). The number of people receiving ART in SSA grew from approximately 5 million in 2010 to about 22 million in 2024. Over that same period, new HIV infections in SSA dropped by 55%, from 1.46 million to 650,000, and AIDS-related deaths dropped by 60%, from 940,000 to 380,000 (4). Despite this progress, the SSA region continues to bear a disproportionate share of the global burden, accounting for 65% of all people living with HIV, 50% of new infections, and 60% of AIDS-related deaths (4).

Key Populations in the HIV Epidemic

Key populations (KPs) are groups of people who, due to social or behavioural factors, have a higher risk of HIV acquisition and higher HIV prevalence. These include sex workers, men who have sex with men (MSM), people who inject drugs (PWID), and people in prisons and other closed settings (8). For example, Korenromp et al. found that the risk of acquiring HIV was 23 times higher for MSM, 14 times higher for PWID, and 8.9 times higher for sex workers, compared with the general adult population (9).

Because KPs have a high burden of HIV, play a disproportionate role in transmission, and face unique structural and societal barriers to healthcare access, effective global management of the HIV epidemic requires prevention and treatment programmes tailored to these populations. A growing body of evidence supports moving care out of traditional hospital settings and into community-based clinics that offer bundled services, including pre-exposure prophylaxis (PrEP) for preventing infection, HIV testing services, concurrent STI screening and treatment, and antiretroviral therapy for those diagnosed with HIV (10,11). There is also support for combining these clinical services with legal and social support structures under so-called “One-Stop Shops”, so that barriers beyond the biomedical, such as violence, discrimination, and legal vulnerability, can be addressed in the same visit (10). However, while these frameworks are widely endorsed, evidence on how they perform at scale remains limited, particularly from large programmatic settings in sub-Saharan Africa, where the HIV burden is greatest.

Key Populations in Nigeria and the Programmatic Response

Nigeria, the most populous country in Africa with over 220 million inhabitants, has the fourth-largest HIV epidemic globally (12,13). Adult HIV prevalence in the 2018 national household survey was 1.3% (14), but the epidemic is heavily concentrated in key populations, where measured prevalence is far higher. The 2020 Integrated Biological and Behavioural Surveillance Survey (IBBSS), which surveyed 17,975 KP members across 12 states, found HIV prevalence of 15.5% among female sex workers (FSWs), 25.0% among MSM, 28.8% among transgender individuals, and 10.9% among PWID, all far exceeding the general population rate (15).

A hostile legal environment for KPs in Nigeria compounds their burden of HIV disease. The Same-Sex Marriage Prohibition Act of 2014 criminalised same-sex relationships with

penalties of up to 14 years' imprisonment (16,17). Drug use is severely penalized, with sentences of up to 25 years' imprisonment for certain substances under the National Drug Law Enforcement Agency Act (18). Sex work occupies a grey legal zone: selling sex is not itself a criminal offence, but soliciting clients and procurement of sex are criminalised under Sections 223–225A of the Criminal Code Act (16,19). Together, these laws push KPs underground, raising barriers to HIV prevention, testing, and treatment (17,20,21).

In response to the concentrated HIV burden among key populations, the Key Population Community HIV Services Action and Response (KP-CARE 1) programme was established in Nigeria with support from USAID and implemented by Heartland Alliance LTD/GTE (22,23). The programme delivers HIV prevention, testing, care, and treatment services for FSWs, MSM, PWID, and transgender individuals across multiple Nigerian states through a differentiated service delivery model centred on community-based One-Stop Shops and mobile outreach (22–24). These One-Stop Shops provide HIV testing and counselling, ART initiation and refills, pre-exposure prophylaxis (PrEP), sexually transmitted infection screening and treatment, and psychosocial support in KP-friendly settings designed to reduce stigma and improve access. In addition, venue-based and hotspot outreach helps link clients from brothels, bars, and other gathering places into prevention and treatment services (11,22,24,25).

Papers 1 and 2 in this thesis use routine client data from KP-CARE 1's electronic data system to analyse outcomes along the HIV care and prevention cascades.

The HIV Care Cascade: Viral Load Testing and Suppression

The HIV Care Cascade is a conceptual model used to monitor the progression of people with HIV through key stages of medical care. It includes testing and diagnosis, linkage to care, initiation of ART, adherence to treatment, viral load testing and viral suppression (26). Attrition happens at each stage of this cascade; understanding these points of attrition is important for evaluating and improving HIV programmes.

Viral suppression is defined as having a very low viral load, commonly < 1000 copies/mL, and is the main goal of ART. Successful viral suppression improves health outcomes and greatly reduces the risk of onward HIV transmission. It is reflected in the UNAIDS 95-95-95 targets, which are global goals targeting a minimum of 95% awareness of status among people living with HIV, 95% ART access among those diagnosed, and 95% viral suppression

among those on ART. In 2024, global estimates were still below these targets, with an estimated 87% aware of their status, 89% of those diagnosed on ART, and 94% of those on ART being virally suppressed (1).

Viral load testing and suppression are especially critical for female sex workers living with HIV because they have a high transmission potential, given their large number of sexual partners. Existing FSW cascade evidence in sub-Saharan Africa is limited and usually is not based on large programme datasets (27).

Paper 1 in this thesis addresses this gap by analysing routine programmatic data from nearly 35,000 FSWs enrolled in the KP-CARE 1 programme to identify factors associated with receipt of viral load testing and achievement of viral suppression.

HIV Prevention: Pre-Exposure Prophylaxis and the Challenge of Continuation

Pre-exposure prophylaxis (PrEP) drugs are antiretroviral medications taken by HIV-negative people to prevent HIV acquisition. The most common PrEP medications consist of daily oral tablets of tenofovir disoproxil fumarate combined with either emtricitabine (FTC) or lamivudine (3TC) (28). Clinical trial and cohort evidence shows that tenofovir-based PrEP can substantially reduce HIV acquisition when taken consistently (29–31).

Despite their proven biological efficacy, the overall impact of PrEP regimens is limited by low uptake among key populations and low continuation after starting. In a global meta-analysis of 59 studies covering over 43,000 participants, 41% had discontinued PrEP within six months of starting, and among those continuing, 37.7% had suboptimal adherence patterns (32). Commonly reported reasons for discontinuation included side effects (25 studies), low perceived risk of infection (21 studies), cost or lack of health insurance (10 studies), and inaccessible care (nine studies). Issues of cost and access are likely to be worse for key populations, who often face criminalisation, stigma, and other barriers to routine health services.

In Nigeria, PrEP rollout for key populations is an important component of the national HIV prevention strategy (33). The KP-CARE 1 programme works to provide PrEP to eligible KP clients using a provider-facilitated approach in which individuals testing HIV-negative are offered oral PrEP as part of a prevention package.

Paper 2 in this thesis examines data from over 43,000 KP clients who initiated PrEP through the KP-CARE 1 programme. The sample size makes it one of the largest PrEP continuation studies among key populations in sub-Saharan Africa. We report the 6-month continuation rates and identify factors associated with sustained PrEP use across different KP groups.

II. DATA SCIENCE CAPACITY BUILDING FOR GLOBAL HEALTH

Bridging Public Health Research and Data Science Capacity

Data analysis skills are increasingly important for the global health workforce. The digitization of health systems, from electronic health records to surveillance and monitoring platforms, generates large volumes of data that can be analysed to improve programme design and delivery, such as in the HIV studies presented in this thesis. Analysing these data sources requires practical data science skills, from cleaning routine datasets and running reproducible analyses to interpreting results for programme use.

These data science skills are still limited in many low- and middle-income countries (LMICs), where they are often needed most. As a result, large amounts of routine health data are under-analysed, and programmes in these countries often rely on external technical support for advanced analyses (34). As a native of Nigeria who has seen these gaps firsthand, helping reduce the shortage of health data professionals in Africa was a core motivation for my doctoral pursuits.

Thus, I have been engaged in building an educational platform and courses, under the banner of The GRAPH Courses, aimed at providing affordable and high-quality educational experiences to help build data capacity among health professionals globally.

Approaches to Data Science Education

Innovative approaches are required to effectively fill the data science skill gap in LMICs. Short in-person professional workshops are one popular format, wherein instructors deliver intensive instruction with hands-on practice, often over a few days to groups of health professionals (35). However, such workshops can be difficult to scale due to their time and financial costs and geographical limitations (36).

Massive open online courses (MOOCs) are another useful education mode for filling the data skills gap. They provide world-class educational content at low or no cost to anyone with access to the internet (37). MOOC platforms like Coursera and edX became very popular in

the 2010s, but enthusiasm for this modality has declined due to their notoriously low completion rates, often in the single digits (38,39). Without a live instructor to provide accountability when student motivation wavers, or support when students get stuck, many learners quickly disengage and drop out (40,41).

Cohort-based online courses are a promising middle ground that blend the flexibility of self-paced courses with the accountability of in-person workshops. These programmes frequently use a flipped classroom model: learners engage with asynchronous content (pre-recorded videos, readings, exercises) on their own schedule, then participate in scheduled live sessions for instructor-led problem-solving and peer interaction (42). The asynchronous component provides flexibility and reduces total contact hours, while the synchronous component provides accountability, community, and direct support, all of which are often absent from purely self-paced courses (43). The rise in remote working during the COVID-19 pandemic created new opportunities to deliver such programmes at scale, because many learners became familiar with the videoconferencing software needed for such sessions (44).

The GRAPH Network and The GRAPH Courses

The GRAPH Network (Global Research and Analyses for Public Health) is an interdisciplinary network of researchers and health professionals headquartered at the University of Geneva's Institute of Global Health. It was established in 2020 in collaboration with the WHO African Regional Office to support the sub-Saharan African COVID-19 response (45). Through the organisation's work with partner countries, we observed firsthand the widespread use of manual processes for data analysis, as few staff had been trained in statistical programming languages, limiting the speed and quality of analyses. Owing partly to this recognition, I proposed and began developing The GRAPH Courses (thegraphcourses.org) as a dedicated training platform for data science and AI in health and life sciences.

The platform delivers training through two complementary models. First, live cohort-based bootcamps use the flipped classroom approach described above: each week, learners complete several hours of asynchronous prework (pre-recorded video lessons, interactive coding exercises, and online quizzes), then attend a 2–3 hour live workshop featuring instructor-led demonstrations, collaborative small-group problem-solving, and peer interaction. The flagship bootcamp is R Basics and Beyond (RBB), which introduces R programming and data analysis for health research and is the subject of the R course

evaluation in this thesis; additional bootcamps cover French-language R, intermediate R, beginner Python, and applied generative AI.

Second, free self-paced versions of these courses—video lessons, exercises, and quizzes without live workshops or instructor interaction—allow learners anywhere in the world to study at their own pace at no cost. Two open-access textbooks, *Introduction to Data Analysis with R* (46) and *Introduction to Data Science with Python* (47), accompany the introductory R and Python courses respectively.

The R course evaluation (Paper 3, in manuscript) examines the educational outcomes of the R Basics and Beyond bootcamp across nine cohorts from 2023 to 2025. Using enrollment rosters and post-course survey data, our study assesses completion rates, participant satisfaction, and learner experience.

Thesis Objectives

This thesis has three objectives:

1. **To examine factors associated with viral load testing and viral suppression among HIV-positive female sex workers initiating antiretroviral therapy in Nigeria** (Paper 1). Using routine programmatic data from nearly 35,000 FSWs enrolled in the KP-CARE 1 programme, this study identifies demographic, clinical, and structural factors associated with receipt of viral load testing and achievement of viral suppression.

2. **To investigate oral PrEP continuation rates and associated factors among key populations in Nigeria** (Paper 2). Using data from over 43,000 KP clients who initiated oral PrEP through the KP-CARE 1 programme, this study examines 6-month continuation rates across different KP groups and identifies factors associated with sustained PrEP use.

3. **To evaluate the educational outcomes of a cohort-based online R programming course for health professionals** (Paper 3). Using enrollment rosters and post-course survey data from nine cohorts of the R Basics and Beyond bootcamp, this study assesses completion rates, participant satisfaction, highlights, and challenges.

Methodological Contributions

Paper 1: Viral Load Testing and Suppression Among FSWs

For Paper 1, I led the statistical analysis, which involved processing and cleaning routine programmatic data from over 34,000 individuals, implementing mixed-effects logistic regression models with random intercepts for 16 care facilities. I produced all figures and tables, wrote the original manuscript draft, and coordinated revisions with co-authors through to publication in *PLoS ONE*.

Paper 2: PrEP Continuation Among Key Populations

For Paper 2, I led the data analysis, including analysing data from over 43,000 PrEP clients, implementing mixed-effects logistic regression with random intercepts for service delivery sites. I decided on the analytical approach in collaboration with the principal investigator, and coordinated with the in-country team on data quality checks (deduplication, logic checks for date and categorical variables). I produced all figures and tables, wrote the original manuscript draft, and coordinated revisions with co-authors through to publication in *BMJ Global Health*.

Paper 3: Evaluation of the R Basics and Beyond Bootcamp

For Paper 3, I was a co-founder and lead developer of The GRAPH Courses initiative, which served as the training platform for the R Basics and Beyond bootcamp. This involved developing the platform's pedagogical approach, the website and learning management system, and the curriculum and content for several bootcamps run by the network.

For the R Basics and Beyond bootcamp, I led the curriculum development and the live workshop facilitation for a majority of the cohorts.

For the evaluation study itself, I designed the post-course survey instrument, led the data analysis, and wrote the manuscript draft.

Papers

The three papers that make up this thesis are included below, arranged thematically rather than in order of publication, as follows:

1. Paper 1: **Viral Load Testing and Suppression Among FSWs**
2. Paper 2: **PrEP Continuation Among Key Populations**
3. Paper 3: **Evaluation of the R Basics and Beyond Bootcamp**

Paper 1: Viral Load Testing and Suppression Among Female Sex Workers

Title: Factors associated with viral load testing and viral suppression among HIV-positive female sex workers in Nigeria

Authors: Nwosu KD, Kalaiwo A, Ngambi WF, Estill J, Omo-Emmanuel UK, Emmanuel G, Keiser O

Journal: *PLoS ONE* 2024; 19(5): e0304487

DOI: 10.1371/journal.pone.0304487

Status: Published, May 2024

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Paper 2: PrEP Continuation Among Key Populations

Title: Oral PrEP continuation rates among key populations in Nigeria: a retrospective cohort study of a large-scale HIV prevention programme

Authors: **Nwosu KD**, Kalaiwo A, Ng'ambi W, Oyelaran O, Umoh P, Estill J, Keiser O

Journal: *BMJ Global Health* 2025; 10(12): e018739

DOI: 10.1136/bmjgh-2024-018739

Status: Published, December 2025

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Paper 3: Evaluation of the R Basics and Beyond Bootcamp

Title: R Basics and Beyond: Outcomes of a Cohort-Based Online Programming Course for Health Professionals

Authors: Nwosu KD (co-authors to be finalized)

Target journal: *Cogent Education*

Status: Manuscript in preparation (draft dated 2026-04-26)

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Conclusions and Perspectives

Summary of Key Findings

The viral suppression analysis (Paper 1) examined viral load testing and suppression among 34,976 HIV-positive female sex workers initiating ART in Nigeria between 2016 and 2022. Both indicators were high: 97.1% received at least one viral load test, and 94.5% were virally suppressed at the test closest to six months post-initiation, approaching the UNAIDS target of 95% suppression among people on ART (1). The odds of both outcomes increased for FSWs initiating ART in more recent years, consistent with improvements in the programme's reach and provision of care over the study period. Advanced WHO clinical stage at baseline was associated with lower odds of suppression, pointing to the importance of early diagnosis and ART initiation.

The PrEP analysis (Paper 2) examined continuation among 43,788 key population clients enrolled across seven Nigerian states between 2020 and 2023. The 6-month continuation rate was 11.5%, substantially lower than comparable estimates of 28% from Cameroon and 25% from Zambia (48,49). Continuation varied markedly across KP groups, with transgender individuals showing the lowest rate (3.5%; aOR 0.37), likely reflecting higher levels of stigma, mistrust of healthcare providers, and competing health priorities in this group. Older clients had higher continuation than younger clients, and unemployment was associated with a small but significant reduction in continuation, suggesting that economic insecurity may impede sustained PrEP use.

The R course evaluation (Paper 3) assessed the R Basics and Beyond bootcamp across nine cohorts of 458 health professionals enrolled between 2023 and 2025. The completion rate of 55.9% substantially exceeded the single-digit rates typical of self-paced MOOCs (38), and 96% of post-course survey respondents gave a recommendation score of 9 or 10 out of 10. Pre-recorded video lessons (mean rating 9.3 out of 10) and live workshop facilitator interaction (9.2) were the highest-rated course components, highlighting the value of both asynchronous content delivery and synchronous cohort interaction. Time constraints (34%) and competing work priorities (30%) were the most frequently cited barriers, underscoring the limitation of a single fixed cohort cadence.

Comparing Outcomes Across the Treatment and Prevention Arms

Considered together, the findings from the viral suppression and PrEP analyses point to contrasting outcomes along the two arms of the KP-CARE 1 cascade. On the treatment side, once FSWs were diagnosed and linked to care, most received viral load testing and achieved suppression, with proportions approaching the UNAIDS 95-95-95 targets (1). These treatment-side results are consistent with evidence that community-based ART delivery for KPs can achieve outcomes at least as good as facility-based care (11), and align with the wider argument that One-Stop Shop and peer-led models can support cascade progression once linkage to care is achieved (10,50).

On the prevention side, the 6-month PrEP continuation rate was considerably lower than estimates from comparable programmes in the region (48,49), and sits at the low end of the continuation rates documented in a global meta-analysis of PrEP studies (32). This finding suggests that, although the provider-facilitated approach has supported broad initial uptake, translating that uptake into sustained use has been more difficult. The drivers of PrEP initiation and the conditions required for sustained use are likely distinct, with continuation shaped by ongoing perceptions of risk, side effects, and the practical demands of daily oral dosing (51,52).

Future Research Directions: Viral Suppression Among FSWs

Beyond the high overall rates of viral load testing and suppression, two avenues for further work emerge from the analysis.

First, the steady improvement in both testing and suppression rates over the study period suggests that the programme's quality improvement efforts have been effective. However, the present analyses cannot identify which programmatic changes drove these gains. Qualitative research, including interviews with service providers and clients, would be well placed to surface the specific shifts in practice that contributed to the improvement. Plausible contributors include the expansion of viral load testing capacity, the transition to more effective ART regimens (notably the shift from efavirenz- to dolutegravir-based regimens), and the refinement of community-based service delivery; documenting these would be valuable for guiding similar programmes in other settings.

Second, the present analysis was limited to outcomes within roughly six months of ART initiation and to indicators measurable in the routine programme dataset. Future analyses

incorporating CD4 counts (which were unavailable here), as well as longer-term outcomes such as retention in care, treatment switching, and mortality, would provide a fuller picture of the care cascade for this population.

Improving PrEP Continuation: Programme and Research Directions

The low PrEP continuation rates observed in the PrEP analysis point to several directions for programme improvement and further research.

First, the optimal balance between breadth of PrEP initiation and depth of pre-initiation engagement remains unresolved. The provider-facilitated approach used in the KP-CARE 1 programme prioritises uptake by offering PrEP to all eligible HIV-negative clients; an alternative would be to incorporate more thorough readiness assessment and motivational counselling at initiation, with the aim of identifying clients more likely to sustain use. More evidence from key population settings is needed to determine which approach is optimal for maximising uptake while remaining cost-effective.

Second, the disparities across KP groups call for group-specific interventions. Transgender individuals, for example, had continuation rates nearly four times lower than FSWs. For transgender individuals, addressing the intersecting challenges of stigma, discrimination, competing health priorities (such as gender-affirming care), and healthcare avoidance may require dedicated service delivery strategies, including transgender-competent providers and peer support (53).

Third, a key limitation of this work is that the routine refill records used could not establish why a client stopped refilling: non-adherence, loss to follow-up, or intentional discontinuation when perceived risk or circumstances changed. Key populations often experience fluctuating HIV risk, and some clients may appropriately cycle on and off PrEP in response to changing risk behaviours (52,54), so the low continuation rate cannot be read automatically as programme failure. Future research should distinguish between discontinuation due to poor adherence and intentional, risk-informed cessation, as these have very different programmatic implications.

Finally, the emergence of long-acting injectable PrEP (cabotegravir, administered every two months (55); lenacapavir, administered every six months (56,57)) and the dapivirine vaginal ring (58) as alternatives to daily oral PrEP may help address some of the barriers to continuation identified in this study. By reducing the burden of daily pill-taking, these newer

modalities may improve continuation rates among key populations who struggle with the daily oral regimen. Lenacapavir introduction in Nigeria has in fact already begun, with rollout commencing in April 2026 (59), and dedicated evaluation will be needed to determine whether it, and the other modalities, deliver this benefit, and for which KP groups.

Building Data Science Capacity: Lessons from the R Bootcamp

Turning to the educational evaluation (Paper 3), the bootcamp findings have implications for the broader design of data science training in global health.

The high ratings of both asynchronous (video lessons; mean 9.3 out of 10) and synchronous (live workshops; 9.2) components support the pedagogical theory underlying the flipped classroom model: that learning can be optimised when self-paced content delivery is paired with guided practice and instructor interaction (42). The particularly high value placed on instructor support during live workshops suggests that for programming education, where learners frequently encounter frustrating errors that are difficult to resolve independently, real-time human guidance remains indispensable. Purely asynchronous formats are likely to under-serve programming learners, while purely synchronous formats forgo the flexibility that allows working professionals to engage with content on their own schedule.

The finding that time constraints (34% of respondents) and competing work priorities (30%) were the primary barriers to completion highlights a fundamental tension in professional education: the working professionals who most need these skills are often those least able to commit sustained time to structured coursework. Notably, non-completing survey respondents reported satisfaction nearly identical to that of completers (mean recommendation score 9.8 in both groups), suggesting that drop-off was driven largely by external time pressures rather than dissatisfaction with the course itself. This has design implications for any data science training aimed at working health professionals: rather than focusing only on improving content, programmes may benefit more from reducing the time-related friction of completion. Possible adaptations include modular formats that allow learners to drop in and out across weeks, built-in catch-up windows that reduce the cost of falling behind, and the offering of multiple tracks at different intensities so learners can match the workload to their capacity.

Completion rates also varied substantially across cohorts, from 45.2% to 70.5%, with no single explanation emerging from the available data. The variation may reflect differences in

cohort composition, in instructional delivery across teaching teams, or in external factors such as timing relative to academic calendars. Identifying the drivers, particularly through more granular monitoring of cohort-level inputs, would help target the programme's quality improvement efforts.

Future Directions for Data Science Training

While the bootcamp results are encouraging, translating them into lasting workforce capacity will require both methodological refinement and broader reach. For the R Basics and Beyond bootcamp itself, longitudinal follow-up studies tracking whether course completion translates into sustained R use and improved work outputs in professional practice would strengthen the evidence base for this training model. Controlled study designs comparing cohort-based and purely self-paced approaches are a valuable future direction.

Beyond the bootcamp evaluated here, The GRAPH Courses platform has, as of early 2026, reached over 9,000 learners from more than 60 countries across its self-paced and cohort-based offerings, with over 500 graduates from the intensive live bootcamps. This reach has been made possible by combining free self-paced courses and video lessons with low-cost live bootcamps. The free self-paced tracks lower the barrier to entry and serve learners who cannot afford tuition or who prefer self-directed study. The platform's YouTube channel (youtube.com/@thegraphcourses) has accumulated over 163,000 views across 113 videos, providing additional open-access educational content. The live bootcamps, supported by a model that combines grant funding with course revenue, allow the platform to maintain low fees and provide substantial financial aid to learners from lower-resource settings. Together, these tracks enable the platform to serve the LMIC populations it aims to reach.

Building on this base, the platform is positioned for continued expansion along several axes. Thematically, the addition of courses in advanced statistics, machine learning, and causal inference would extend the curriculum beyond introductory data skills into the more specialised competencies increasingly demanded by global health research. Institutionally, the hosting of additional external courses from partner organisations, building on the existing collaborations with WHO and the University of Oxford, offers the potential to establish the platform as a shared infrastructure for data science training across the global health sector.

Conclusion

The three studies in this thesis are united by the premise that data-driven decision-making can improve health outcomes for vulnerable populations, but only if both the data and the skills to analyse it are in place. Papers 1 and 2 demonstrate this premise in action: routine programmatic data, subjected to rigorous analysis, showed that the KP-CARE 1 programme is achieving strong treatment outcomes for FSWs on the care cascade, while also revealing a critical gap in PrEP continuation, particularly among transgender individuals and younger clients. These findings can directly inform programme design, but they required analytical skills that remain scarce in the settings where such programmes operate. Moreover, rigorous interpretation of programmatic data depends on contextual knowledge of the system that produced it, meaning that in-country analysts, who already hold that knowledge, are often best placed to generate high-quality evidence. Building local data science capacity would therefore improve both the quantity and quality of the resulting evidence.

The R Basics and Beyond bootcamp examined in Paper 3 represents one approach to addressing this scarcity, and its evaluation suggests that cohort-based online training is a viable model for doing so at scale.

Ultimately, this thesis argues that strengthening the HIV response for key populations, and improving global health more broadly, depends not only on better interventions and better data, but on expanding who has the ability to turn data into evidence.

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